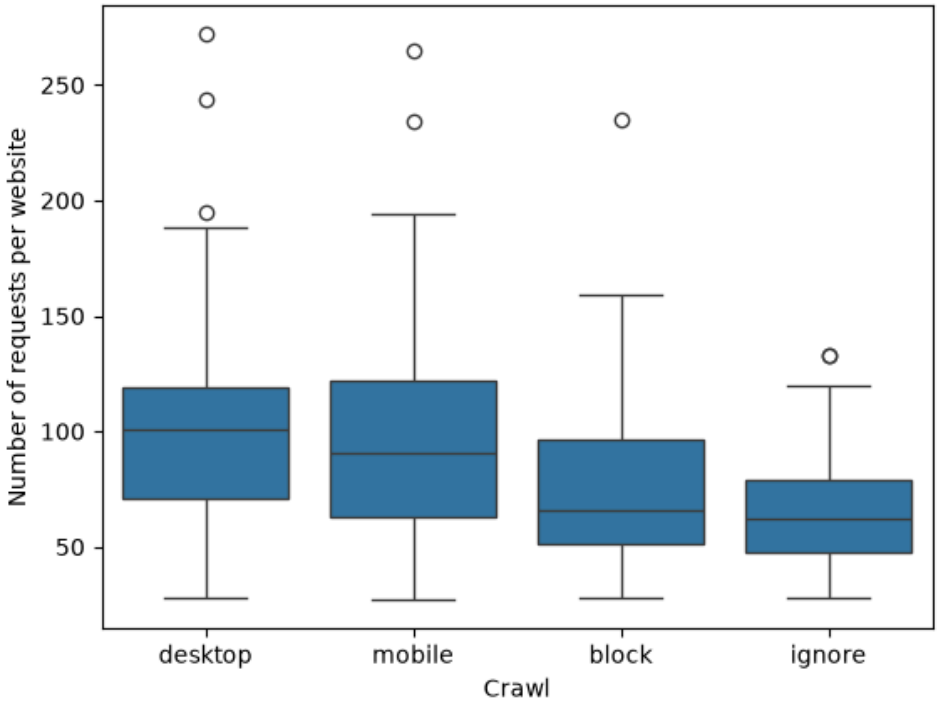
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Online Tracking and Privacy
Assignment 2

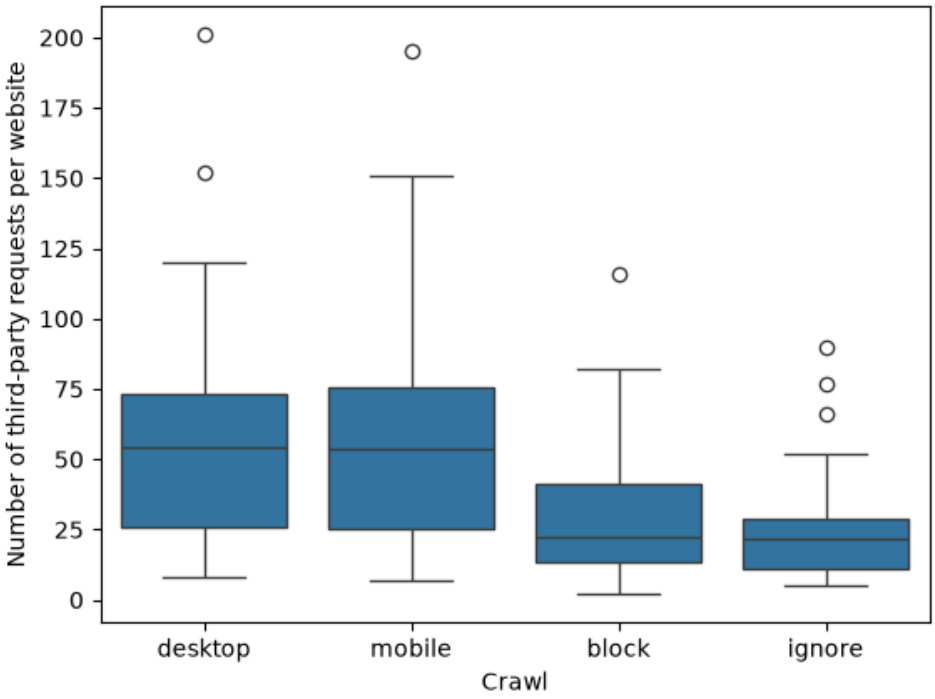
Group A2 11

Analysis:

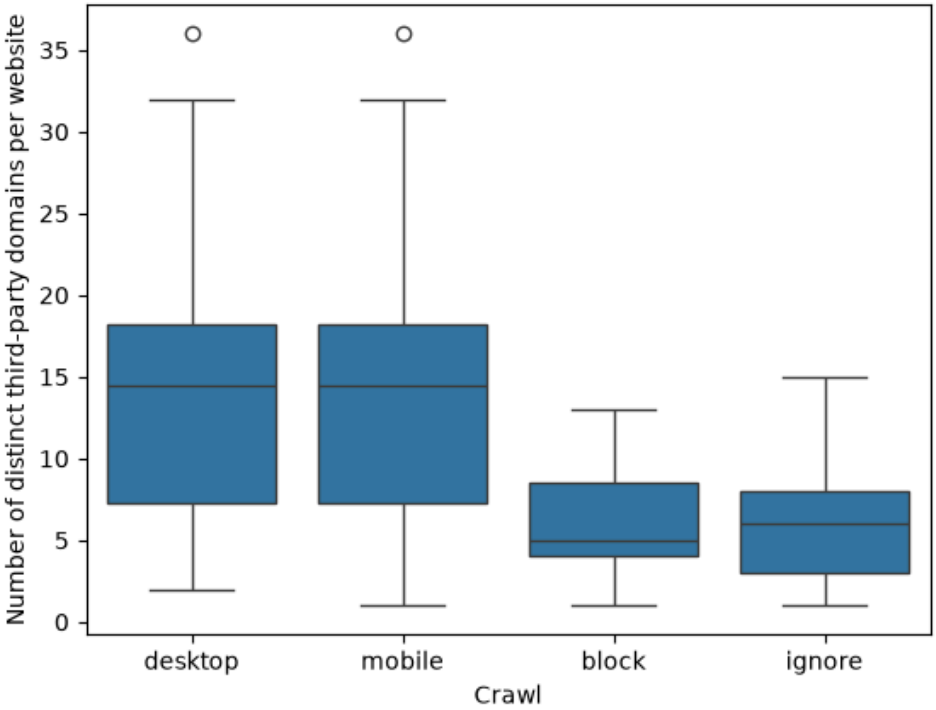
1. Box plots
    a. Number of requests per website



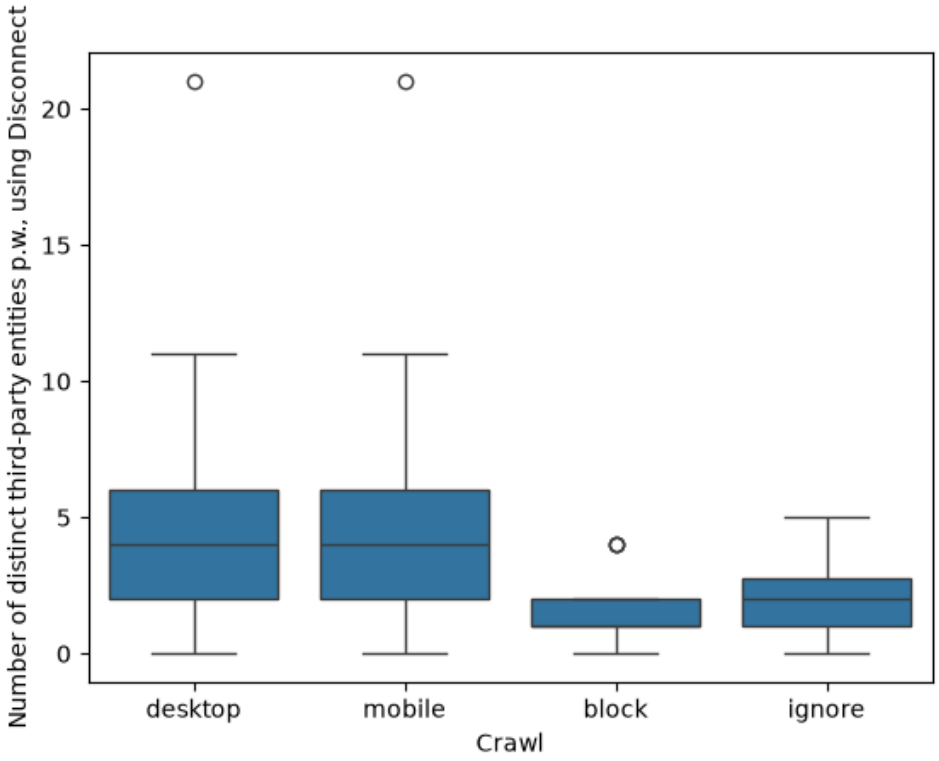
b. Number of third-party requests per website



c. Number of distinct third-party domains per website



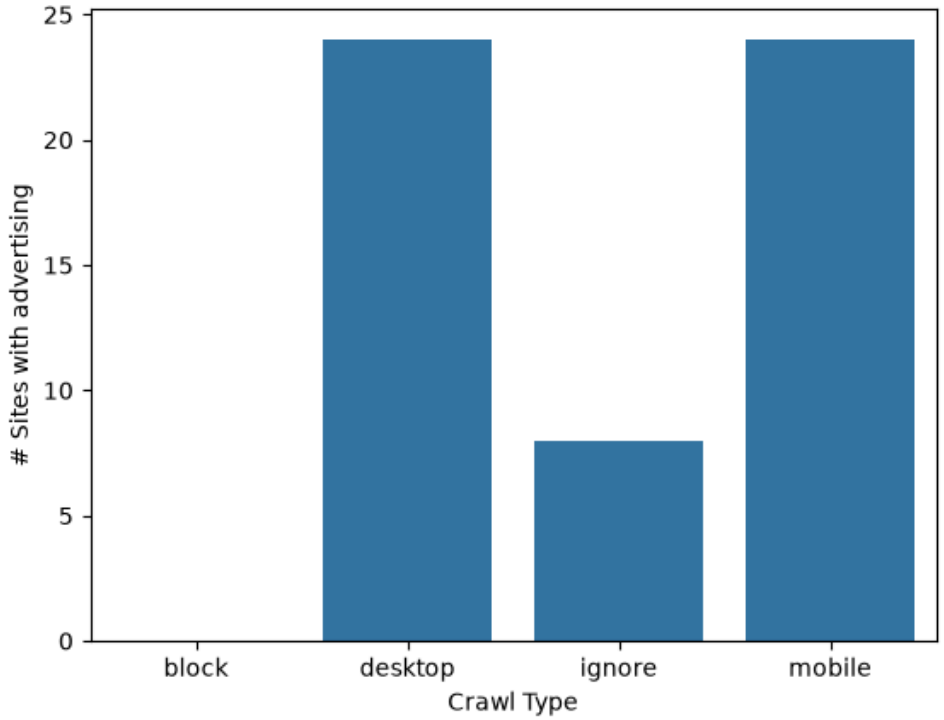
d. Number of distinct third-party entities per website, using Disconnect's entity list



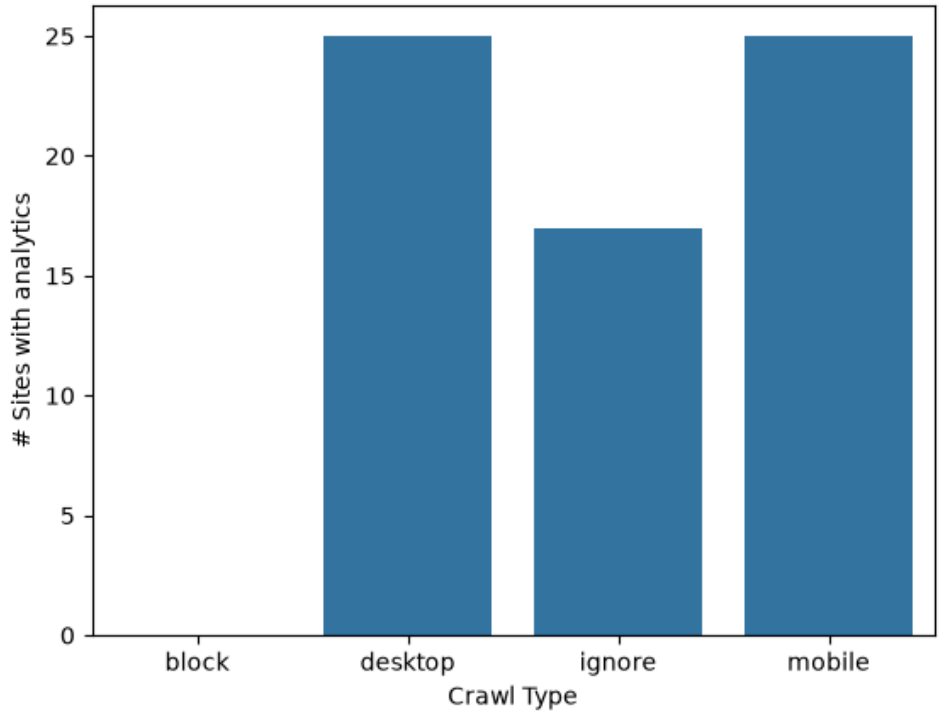
2. Data comparison table from 1a) - 1d)

mode	block			desktop			ignore			mobile		
	min	median	max	min	median	max	min	median	max	min	median	max
num_requests	28.0	66.0	235.0	28.0	101.0	272.0	28.0	62.0	133.0	27.0	90.5	265.0
num_third_party_requests	2.0	22.0	116.0	8.0	54.5	201.0	5.0	21.5	90.0	7.0	53.5	195.0
num_dist_third_party_domains	1.0	5.0	13.0	2.0	14.5	36.0	1.0	6.0	15.0	1.0	14.5	36.0
num_dist_third_party_entites_per_website	0.0	1.0	4.0	0.0	4.0	21.0	0.0	2.0	5.0	0.0	4.0	21.0

3. Bar plots
    a. Websites that send **at least one request** to a domain categorized as *Advertising* in Disconnect's blocklist



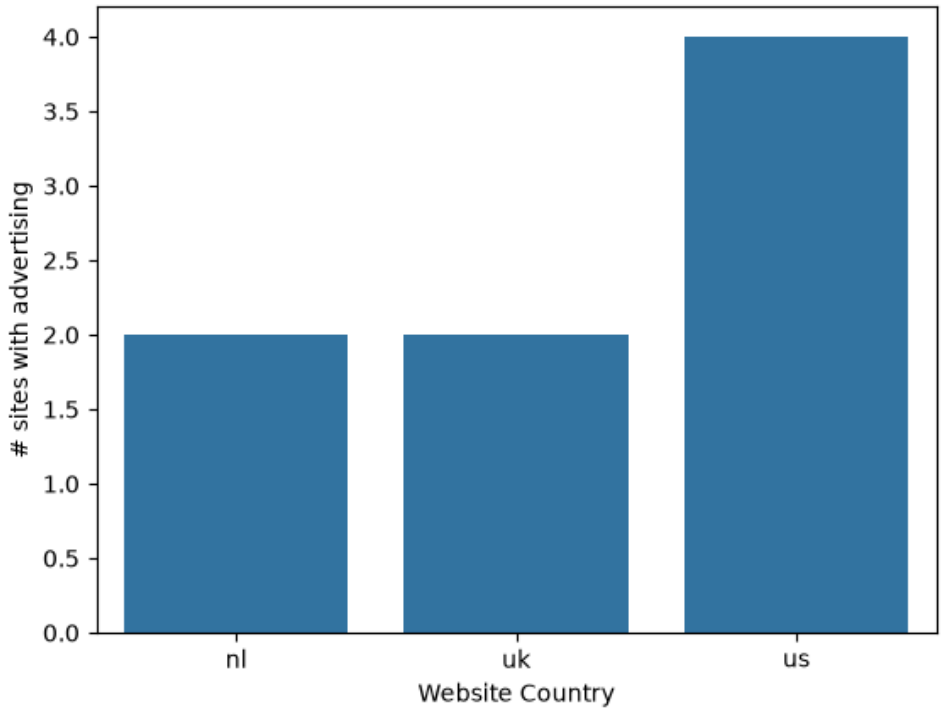
    b. Websites that send **at least one request** to a domain categorized as *Analytics* in Disconnect's blocklist



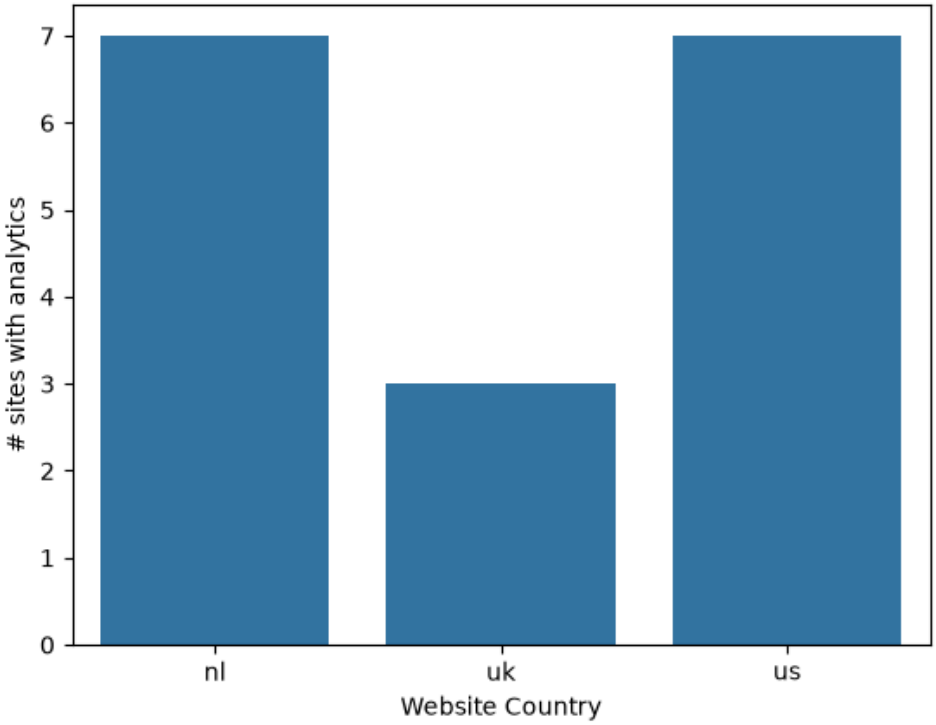
4. Data comparison table from 1a) - 1d) based on countries

country	nl			uk			us		
	min	median	max	min	median	max	min	median	max
num_requests	28.0	69.0	272.0	85.0	143.5	244.0	28.0	82.5	120.0
num_third_party_requests	9.0	37.5	74.0	50.0	105.0	201.0	8.0	27.0	77.0
num_dist_third_party_domains	3.0	10.5	19.0	15.0	22.0	36.0	2.0	8.5	15.0
num_dist_third_party_entites_per_website	1.0	4.5	8.0	2.0	7.5	21.0	0.0	2.0	5.0

5. Bar plots
   a. Number of websites from each country that send **at least one request** to an *Advertising* domain in Disconnect's blocklist from the Ignore crawl



b. Number of websites from each country that send **at least one request** to an *Analytics* domain in Disconnect's blocklist from the Ignore crawl



6. The three most prevalent third-party domains contacted uniquely in the Desktop crawl (not in Mobile) and uniquely in the Mobile crawl (not in Desktop):

	Domains observed in Crawl-Desktop		Domains observed in Crawl-Mobile	
	Domain	**Site count**	**Domain**	**Site count**
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-

Short comment: The results show that the Desktop and the Mobile crawls contacted the same 136 third-party domains.

7. The ten most prevalent third-party domains, based on the number of distinct websites where the third-party domain was contacted, as well as the categories from the Disconnect blocklist.

a. Desktop

	Third-party domain	Number of distinct websites	Disconnect categories
1	googletagmanager.com	28	NA
2	google.com	22	Content
3	doubleclick.net	22	Advertising, Email, FingerprintingGeneral
4	google.nl	21	Content
5	google-analytics.com	16	Analytics, Email, FingerprintingGeneral
6	facebook.net	14	FingerprintingGeneral, Social
7	facebook.com	14	FingerprintingGeneral, Social
8	linkedin.com	13	Social
9	licdn.com	13	Social
10	siteimproveanalytics.io	9	Analytics

b. Mobile

	Third-party domain	Number of distinct websites	Disconnect categories
1	googletagmanager.com	27	NA
2	google.com	22	Content
3	doubleclick.net	22	Advertising, Email, FingerprintingGeneral
4	google.nl	21	Content
5	google-analytics.com	16	Analytics, Email, FingerprintingGeneral
6	facebook.net	14	FingerprintingGeneral, Social
7	facebook.com	14	FingerprintingGeneral, Social
8	linkedin.com	13	Social
9	licdn.com	13	Social
10	siteimproveanalytics.io	9	Analytics

c. Block

	Third-party domain	Number of distinct websites	Disconnect categories
1	googletagmanager.com	28	NA
2	google.nl	18	Content
3	gstatic.com	9	Content
4	google.com	8	Content
5	fonts.googleapis.com	8	NA
6	youtube.com	7	Content
7	cloudflare.com	6	Content
8	ytimg.com	5	Content
9	onetrust.com	5	ConsentManagers
10	typekit.net	4	NA

d. Ignore

	Third-party domain	Number of distinct websites	Disconnect categories
1	googletagmanager.com	23	NA
2	google-analytics.com	11	Analytics, Email, FingerprintingGeneral
3	gstatic.com	8	Content
4	google.com	7	Content
5	fonts.googleapis.com	7	NA
6	cloudflare.com	6	Content
7	youtube.com	5	Content
8	siteimproveanalytics.io	5	Analytics
9	siteimproveanalytics.com	5	Analytics
10	doubleclick.net	5	Advertising, Email, FingerprintingGeneral

8. Number of distinct third-party domains in the Desktop crawl, including the website's Tranco rank (Tranco rank retrieved 10 November 2026)

a. Highest number of distinct third-party domains

	Website	Tranco rank	Country	Number of distinct third-party domains
1	nottingham.ac.uk	8540	UK	36
2	leeds.ac.uk	5952	UK	32
3	manchester.ac.uk	4951	UK	31

b. Lowest number of distinct third-party domains

	Website	Tranco rank	Country	Number of distinct third-party domains
1	mit.edu	238	US	2
2	rug.nl	5587	NL	3
3	utwente.nl	6529	NL	4

9. All websites where a session replay domain was contacted in one or more crawls

	Website	Session replay domain	Country	Observed crawl(s)
1	lse.ac.uk	hotjar.com	UK	Desktop, Mobile
2	lse.ac.uk	hotjar.io	UK	Desktop, Mobile
3	manchester.ac.uk	hotjar.com	UK	Desktop, Mobile
4	stanford.edu	contentsquare.net	US	Desktop, Mobile, Ignore
5	uchicago.edu	hotjar.com	US	Desktop, Mobile, Ignore
6	uchicago.edu	hotjar.io	US	Mobile
7	warwick.ac.uk	hotjar.com	UK	Desktop, Mobile

10. Five permissions that are explicitly disabled in the Mobile crawl

	Permission name	Number of distinct websites where this permission was disabled
0	microphone	5
1	camera	4
2	gyroscope	3
3	magnetometer	3
4	payment	2

11. Websites where a no-referrer Referrer Policy was set by a first-party response in the Desktop crawl

	Website	Country
0	eur.nl	NL
1	utwente.nl	NL
2	uva.nl	NL

12. Requests with first-party subdomain, CNAME record, CNAME entity and Disconnect categories

a. Desktop

	Website	First-party subdomain	CNAME record	CNAME entity	Disconnect categories
0	manchester.ac.uk	www.manchester.ac.uk	duc7h5khrpi4j.cloudfront.net	Amazon	Content
1	nottingham.ac.uk	smetrics.nottingham.ac.uk	oacflwpnfh.data.adobedc.net	Adobe	Analytics
2	uchicago.edu	www.uchicago.edu	hubba882888a582-prod-frontdoor-endpoint-gkbrer...	Microsoft	Content
3	ucla.edu	cdn.webcomponents.ucla.edu	d3nstw201xsr54.cloudfront.net	Amazon	Content
4	ucla.edu	www.ucla.edu	d1zev4mn1zpfbc.cloudfront.net	Amazon	Content

b. Mobile

	Website	First-party subdomain	CNAME record	CNAME entity	Disconnect categories
0	manchester.ac.uk	www.manchester.ac.uk	duc7h5khrpi4j.cloudfront.net	Amazon	Content
1	nottingham.ac.uk	smetrics.nottingham.ac.uk	oacflwpnfh.data.adobedc.net	Adobe	Analytics
2	uchicago.edu	www.uchicago.edu	hubba882888a582-prod-frontdoor-endpoint-gkbrer...	Microsoft	Content
3	ucla.edu	cdn.webcomponents.ucla.edu	d3nstw201xsr54.cloudfront.net	Amazon	Content
4	ucla.edu	www.ucla.edu	d1zev4mn1zpfbc.cloudfront.net	Amazon	Content

c. Block

	Website	First-party subdomain	CNAME record	CNAME entity	Disconnect categories
0	manchester.ac.uk	www.manchester.ac.uk	duc7h5khrpi4j.cloudfront.net	Amazon	Content
1	nottingham.ac.uk	smetrics.nottingham.ac.uk	oacflwpnfh.data.adobedc.net	Adobe	Analytics
2	uchicago.edu	www.uchicago.edu	hubba882888a582-prod-frontdoor-endpoint-gkbrer...	Microsoft	Content
3	ucla.edu	cdn.webcomponents.ucla.edu	d3nstw201xsr54.cloudfront.net	Amazon	Content
4	ucla.edu	www.ucla.edu	d1zev4mn1zpfbc.cloudfront.net	Amazon	Content

d. Ignore

	Website	First-party subdomain	CNAME record	CNAME entity	Disconnect categories
0	manchester.ac.uk	www.manchester.ac.uk	duc7h5khrpi4j.cloudfront.net	Amazon	Content
1	uchicago.edu	www.uchicago.edu	hubba882888a582-prod-frontdoor-endpoint-gkbrer...	Microsoft	Content
2	ucla.edu	cdn.webcomponents.ucla.edu	d3nstw201xsr54.cloudfront.net	Amazon	Content
3	ucla.edu	www.ucla.edu	d1zev4mn1zpfbc.cloudfront.net	Amazon	Content

13. Questions:

a. What was the most interesting difference you observed between the four crawls? Refer to at least one specific finding from your analysis.
Consent matters more than anything else in our data. Without interacting with the banner (Ignore), the average website contacted 6 distinct third-party domains, compared with 14.5 after "Accept all" (Desktop). Blocking Disconnect's tracker categories while still accepting (Block) brought this down to 5. The CNAME analysis (Q12) shows the limits of domain-based blocking: on nottingham.ac.uk the first-party subdomain *smetrics.nottingham.ac.uk* points via CNAME to Adobe's analytics endpoint (**.data.adobedc.net*). It was still contacted in the Block crawl because its name looks first-party.

b. Identify a limitation of your crawler or measurement methodology. How might it have affected your results?
A limitation we found was that, when visiting the Radboud University website (ru.nl), the crawler failed to connect to the website. When investigating, a group member used *curl -I https://www.ru.nl*, and it still failed to connect to the website. This suggests that the Radboud website may have a limit for automated visits. Websites like this may have a system that blocks crawlers after a number of visits and it could have affected our results by giving us insufficient data in the form of an incomplete HAR file or no HAR file at all. When running the crawler on a different day, the crawler was successful in retrieving data from the Radboud website.

Another limitation we found was that we did the DNS lookups for Q12 during the analysis (as recommended), not during the crawl, and from a single resolver. If a

CNAME record changed between the crawl and the analysis, or depends on the resolver's location, Q12 could report trackers that did not exist at crawl time. We also crawled each site once, from one vantage point (NL), so geo-dependent consent banners are not captured.

c. What was the most challenging part of the assignment, and how did your group address it?
   The most challenging part of the assignment was creating the crawler algorithm. All parts of the crawler were built upon each other and were interconnected. We solved this by having every group member create their own crawler then choosing the best one. This may have been inefficient but relevant as it gave knowledge about the crawler and how it works for all group members.